NFDI4ING

Task Area Betty — Revised Workflow

Benchmarking Workflows for Engineering Simulation Software

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Version 1: External Communication

The Geschäftsstelle requests a condensed overview for external communication:

- **A clearly named starting point** (e.g., experiment, data collection, problem statement)
- **A clearly defined end station** (e.g., publication, transfer product, consulting result)
- **2–4 central stations in between** that represent the truly decisive steps
- **Conscious prioritization** — which stations are indispensable to make the path from A to Z understandable?
- **Clear reference to core offerings** — the chosen stations should make these visible, not just describe generic process steps

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VERSION 1

External Communication

START Engineering research question (e.g., "Does my simulation software produce correct results?")

END Published, reproducible benchmark results

1 Plan

Define benchmark scope and create a Software Management Plan.

RDMO (Core Service) — create and manage the Software Management Plan

2 Develop & Prepare

Search for benchmarks and simulation software. Develop or adapt simulation software. Define benchmark cases using standardized terminology.

Terminology Service (Core Service) — standardized vocabularies for benchmark descriptions

BRE — search for benchmark descriptions and simulation software

GitLab — version control for simulation code

CIM — standardized schema for benchmark cases

3 Execute & Evaluate

Run benchmarks in a reproducible environment. Compare results against metrics. Create publication-ready figures.

Jupyter Service (Core Service) — reproducible environment for simulations and analysis

fieldcompare — quantitative comparison of field data

gridformat — mesh/grid data format conversion

plotID — traceable, publication-ready plots

4 Publish

Archive research data. Publish article. Link software, data, and benchmarks as FAIR Digital Objects.

ing.grid (Core Service) — Diamond Open Access journal for FAIR publication

Coscine (Core Service) — FAIR data archiving with metadata and PIDs

RO Hub — package outputs as Research Objects

Version 2: Internal Consolidation

For the internal consolidation, a more detailed representation of the overall process is required:

- **A logically structured sequence of stations**
- **Clear and precise naming** of the central steps
- **A clearly recognizable reference of each station to the core offerings**
- **A clearly defined start and end point** of the process
- **Additional stations and intermediate steps** may be included to make the full process visible

Specifically requested improvements:

→ Stronger reference to core offerings
→ Clearer naming of central stations
→ Unambiguous structuring of the process from start to goal

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Personas

SW Developer

Develops or maintains simulation software (e.g., a CFD solver or structural analysis tool). Responsible for code quality, licensing, version control, and publishing the software as a FAIR Digital Object.

BM Developer

Designs and defines benchmark cases for simulation software, including input data, boundary conditions, reference solutions, and evaluation metrics. Ensures benchmarks are well-documented and reproducible.

V&V Engineer

Performs verification and validation by executing benchmarks against simulation software, comparing results to reference data, and assessing whether the software meets defined accuracy criteria.

SW User

Uses existing simulation software and published benchmarks to answer a specific research question. Searches for suitable tools and benchmark cases, runs simulations, and interprets results.

VERSION 2 — INTERNAL

PERSONAS: SW Dev · BM Dev · V&V · User

Full Workflow

START Idea / Research Question

END Published & linked FAIR Digital Objects (software, data, benchmark, publication)

1 Plan

Define research question, create SMP.

RDMO (Core) — create Software Management Plan

OUTPUT Software Management Plan (SMP)

2 Search

Find simulation software, benchmark descriptions, reference data, standardized terminology.

Terminology Service (Core) — standardized vocabularies/ontologies
BRE — search benchmarks & software

OUTPUT Identified software, benchmark descriptions, reference data

3 Develop

Develop/adapt simulation software. Define benchmark cases (input, boundary conditions, metrics) using standardized models.

Terminology Service (Core) — standardized vocabulary for descriptions
GitLab — version control for simulation code
CIM — standardized benchmark description schema
License Checker — verify licensing compliance
Software FDO — register software as FAIR Digital Object

OUTPUT Simulation software (versioned), benchmark description

4 Prepare & Execute

Combine simulation software with benchmark input files. Run simulations in reproducible environment.

Jupyter Service (Core) — reproducible computational environment
gridformat — mesh/grid format conversion

OUTPUT Simulation result files (field data, logs, convergence data)

5 Visualize & Compare

Compare results against benchmark metrics. Create publication-ready figures.

Jupyter Service (Core) — interactive analysis & visualization
fieldcompare — quantitative field data comparison
plotID — traceable publication-ready plots

OUTPUT Figures, benchmark metric evaluations, comparison reports

6 Store & Document

Archive all outputs with FAIR metadata and PIDs. Package as Research Objects.

Coscine (Core) — FAIR storage, metadata, PIDs
RO Hub — bundle outputs as RO-Crate

OUTPUT Archived data with PIDs, Research Object RO Crates

7 Publish & Link

Publish article. Link all outputs into connected FAIR Digital Objects.

ing.grid (Core) — Diamond OA journal, open peer review
Coscine (Core) — persistent linking via PIDs
RO Hub — navigable Research Object
CIM — machine-readable benchmark description

OUTPUT Published article, linked FAIR Digital Objects